How-To Library

Informatica

Importing Metadata From Hive Sources in Test Data Management

© Copyright Informatica LLC 2019, 2021. Informatica, the Informatica logo, and PowerCenter are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html

Abstract

You can perform Test Data Management (TDM) operations on Hive and Hadoop Distributed File System (HDFS) data sources in a Hadoop environment in Test Data Manager.

You must configure a Hadoop environment before you can create the connections in Test Data Manager. You can then import metadata from the source connections to perform TDM operations.

The article describes how to configure a Hadoop environment, configure the Data Integration Service, configure Hive, Hadoop, and HDFS connections, and import metadata from a Hive connection. This article assumes that you have knowledge of TDM and are familiar with TDM operations.

Supported Versions

- Test Data Management 10.2.1
- Test Data Management 10.2.0
- Test Data Management 10.1.0

Table of Contents

Overview. 2
Rules and Guidelines. 3
Installing Informatica Big Data Management RPM. 3
Run the Big Data Management Configuration Utility. 4
Use Cloudera CDH. 4
Use Apache Ambari. 7
Use SSH. 9
Verify Entry Files 11
Copy the Highgroup.txt File to the Hadoop Machine. 11
Import Connections in to Test Data Manager. 11
Create Connections in Test Data Manager. 11
Configure Pushdown Connection Properties in Test Data Manager. 12
Create a Project and Import Source Metadata. 13

Overview

You can perform TDM operations on data from Hadoop clusters. To perform TDM operations on source data you must import the source metadata into the TDM repository.

Create connections in Test Data Manager to import metadata and perform TDM operations on the source metadata.

You can create the following connections in Test Data Manager to perform TDM operations on data in Hadoop clusters:

- Hive. Use as a source or target connection in a plan.
- HDFS. Use as a source or target connection in a plan.
- Hadoop. Required as a pushdown connection.

Perform the following steps to import source metadata into Test Data Manager:

1. Download and install the Informatica Big Data Management installation package.
2. Run the Big Data Management Configuration Utility. The utility performs the following tasks:
   - Creates configuration files on the machine where the Data Integration Service runs.
   - Creates connections between the cluster and the Data Integration Service.
   - Updates Data Integration Service properties in preparation for running mappings on the cluster.
3. Verify the ODBC entry files.
4. Optional. Copy the `highgroup.txt` file to the Hadoop machine. Required if you want to perform SSN masking.
5. Import connections in to Test Data Manager.
6. Optional. Create connections in Test Data Manager.
7. Configure pushdown connection properties in Test Data Manager.
8. Create a project and import metadata from a Hive connection in Test Data Manager.

Rules and Guidelines

When you create connections to work with data in Hadoop clusters in a TDM operation, consider the following rules and guidelines:

- You must create a Hadoop connection to use as a pushdown connection in workflows that contain Hive or HDFS connections.
- You cannot use a Hadoop connection for data movement.
- You can import source metadata from a Hive connection into TDM.
- You can import source metadata from a HDFS source from PowerCenter.

  Note: This document does not include the steps to import source metadata from PowerCenter.

Installing Informatica Big Data Management RPM

You can install the RPM package on a single node or in a cluster.

1. Download the following file to a temporary folder: `InformaticaHadoop-<InformaticaForHadoopVersion>.tar.gz`.

   The distribution package must be stored on a local disk and not on HDFS.
2. Extract the file to the machine from where you want to distribute the package and run the Big Data Management installation.
3. Copy the `InformaticaHadoop-<InformaticaForHadoopVersion>.rpm` to a shared directory based on the transfer protocol you are using.

   The file location must be accessible to all the nodes in the cluster.
4. Run the following command from the Big Data Management root directory to start the installation in console mode:

   bash InformaticaHadoopInstall.sh

   For more information about installation, see the *Informatica Big Data Management 10.1 Installation and Configuration Guide*. The installer creates the `<Big Data Management installation directory>/Informatica` directory and populates all of the file systems with the contents of the RPM package.

5. In a cluster environment, you must install Big Data Management on each node in the cluster.

   To get more information about the tasks performed by the installer, you can view the `Informatica-hadoop-install.<DateTimeStamp>.log` installation log file.

Run the Big Data Management Configuration Utility

You can use the Big Data Management Configuration Utility to automate part of the configuration for Big Data Management. After you run the utility, complete the configuration process for Big Data Management.

Note: The utility does not support Big Data Management for the following distributions:

- Highly available BigInsights clusters
- Amazon EMR
- Azure HDInsight
- MapR

1. On the machine where the Data Integration Service runs, open the command line.
2. Go to the following directory: `<Informatica installation directory>/tools/BDMUtil`.
3. Run `BDMConfig.sh`.
4. Press `Enter`.
5. Choose the Hadoop distribution that you want to use to configure Big Data Management:

Option	Description
1	Cloudera CDH
2	Hortonworks HDP
3	MapR
4	IBM BigInsights

   Note: Select only 1 for Cloudera or 2 for Hortonworks. At this time, the utility does not support configuration for MapR or BigInsights.

6. Based on the option you selected in step 5, see the corresponding topic to continue with the configuration process.

Use Cloudera CDH

If you choose Cloudera CDH, perform the following steps to configure Big Data Management:

1. Select the version of Cloudera CDH to configure.
2. Choose the method to access files on the Hadoop cluster:

   The following options appear:

Option	Description
1	Cloudera Manager. Select this option to use the Cloudera Manager API to access files on the Hadoop cluster.

Option	Description
2	Secure Shell (SSH). Select this option to use SSH to access files on the Hadoop cluster. This option requires SSH connections to the machines that host the name node, JobTracker, and Hive client. If you select this option, Informatica recommends that you use an SSH connection without a password or have sshpass or Expect installed.

Note: Informatica recommends the Cloudera Manager option.

3. Select whether to use HiveServer 2 to run mappings.

   Note: If you select no, Big Data Management uses the default Hive driver to run mappings.

4. Verify the location of the Informatica Big Data Management installation directory on the cluster.

   The default location appears, along with the following options:

Option	Description
1	OK. Select this option to change the directory location.
2	Continue. Select this option to accept the default directory location.

5. Configure the connection to the Cloudera Manager.
   a. Enter the Cloudera Manager host.
   b. Enter the Cloudera user ID.
   c. Enter the password for the user ID.
   d. Enter the port for Cloudera Manager.
   e. Select whether to use Tez as the execution engine type.
      - 1 - No
      - 2 - Yes

   The Big Data Management Configuration Utility retrieves the required information from the Hadoop cluster.

6. Select whether to exit the utility, or update the Data Integration Service and create connections.

   Select from the following options:

Option	Description
1	No. Select this option to exit the utility.
2	Yes. Select this option to continue.

7. Select whether to update Data Integration Service properties.

   Select from the following options:

Option	Description
1	No. Select this option to update Data Integration Service properties later.
2	Yes. Select this option to update Data Integration Service properties now.

8. Choose if you want to create connections to the Hadoop cluster:

Option	Description
1. Hadoop	Create a Hadoop connection.
2. Hive	Create a Hive connection.
3. HDFS	Create an HDFS connection.
4. HBase	Create an HBase connection.
5. Select all	Create all four types of connection.

   Press the number that corresponds to your choice.

9. Supply information about the Informatica domain.
   a. Enter the domain name.
   b. Enter the node name.
   c. Enter the domain user name.
   d. Enter the domain password.
   e. Enter the Data Integration Service name.
   f. If the cluster is Kerberos-enabled, enter the following additional information:
      - Hadoop kerberos service principal name
      - Hadoop kerberos keytab location -- Location of the keytab on the Data Integration Service machine.

   Note: After you enter the Data Integration Service name, the utility tests the domain connection, and then recycles the Data Integration Service.

10. In the **Connection Details** section, provide the connection properties.

    Based on the type of connection you choose to create, the utility requires different properties. For more information about the connection properties, see the *Big Data Management User Guide*.

    Note: When you specify a directory path for the Blaze working directory or the Spark staging directory, you must specify existing directories. The Big Data management utility does not validate the directory paths that you specify.

    The utility creates the connections that you configured.

11. The utility reports a summary of its operations, including whether connection creation succeeded, and the location of utilty log files.
12. Complete the manual configuration steps for Big Data Management.

The utility creates the following files in the <Informatica installation directory>/tools/BDMUtil directory:

ClusterConfig.properties

Contains details about the properties fetched from the Hadoop cluster and connection creation commands that can be used to create connections to the Hadoop cluster.

Note: Edit the connection name, domain username and password to use the generated commands.

HiveServer2_EnvInfa.txt

Contains the list of environment variables and values that need to be copied to the HiveServer2 environment on the Hadoop cluster. This file is created only if you choose HiveServer2.

Use Apache Ambari

If you choose Ambari, perform the following steps to configure Big Data Management:

1. Select the Hadoop distribution directory to configure.
2. Choose the method to access files on the Hadoop cluster:

   The following options appear:

Option	Description
1	Apache Ambari. Select this option to use the Ambari REST API to access files on the Hadoop cluster.
2	Secure Shell (SSH). Select this option to use SSH to access files on the Hadoop cluster. This option requires SSH connections to the machines that host the name node, JobTracker, and Hive client. If you select this option, Informatica recommends that you use an SSH connection without a password or have sshpass or Expect installed.

   Note: Informatica recommends the Apache Ambari option.
3. Select whether to use HiveServer 2 to run mappings.

   Note: If you select no, Big Data Management uses the default Hive Command Line Interface to run mappings.
4. Verify the location of the Informatica Big Data Management installation directory on the cluster.

   The default location appears, along with the following options:

Option	Description
1	OK. Select this option to change the directory location.
2	Continue. Select this option to accept the default directory location.

5. Configure the connection to the Ambari Manager.
   a. Enter the Ambari Manager host.
   b. Enter the Ambari user ID.
   c. Enter the password for the user ID.
   d. Enter the port for Ambari Manager.
   e. Select whether to use Tez as the execution engine type.
      - 1 - No
      - 2 - Yes

   The Big Data Management Configuration Utility retrieves the required information from the Hadoop cluster.
6. Select whether to exit the utility, or update the Data Integration Service and create connections.

   Select from the following options:

Option	Description
1	No. Select this option to exit the utility.
2	Yes. Select this option to continue.

7. Select whether to update Data Integration Service properties.

Select from the following options:

Option	Description
1	No. Select this option to update Data Integration Service properties later.
2	Yes. Select this option to update Data Integration Service properties now.

8. Choose if you want to create connections to the Hadoop cluster:

Option	Description
1. Hadoop	Create a Hadoop connection.
2. Hive	Create a Hive connection.
3. HDFS	Create an HDFS connection.
4. HBase	Create an HBase connection.
5. Select all	Create all four types of connection.

Press the number that corresponds to your choice.

9. Supply information about the Informatica domain.
   a. Enter the domain name.
   b. Enter the node name.
   c. Enter the domain user name.
   d. Enter the domain password.
   e. Enter the Data Integration Service name.
   f. If the cluster is Kerberos-enabled, enter the following additional information:
      - Hadoop kerberos service principal name
      - Hadoop kerberos keytab location -- Location of the keytab on the Data Integration Service machine.

   Note: After you enter the Data Integration Service name, the utility tests the domain connection, and then recycles the Data Integration Service.

10. In the **Connection Details** section, provide the connection properties.

    Based on the type of connection you choose to create, the utility requires different properties. For more information about the connection properties, see the *Big Data Management User Guide*.

    Note: When you specify a directory path for the Blaze working directory or the Spark staging directory, you must specify existing directories. The Big Data management utility does not validate the directory paths that you specify.

    The utility creates the connections that you configured.

11. The utility reports a summary of its operations, including whether connection creation succeeded, and the location of utilty log files.
12. Complete the manual configuration steps for Big Data Management.

The utility creates the following files in the <Informatica installation directory>/tools/BDMUtil directory:

ClusterConfig.properties

Contains details about the properties fetched from the Hadoop cluster and connection creation commands that can be used to create connections to the Hadoop cluster.

Note: Edit the connection name, domain username and password to use the generated commands.

HiveServer2_EnvInfa.txt

Contains the list of environment variables and values that need to be copied to the HiveServer2 environment on the Hadoop cluster. This file is created only if you choose HiveServer2.

Use SSH

If you choose SSH, you must provide host names and Hadoop configuration file locations.

Note: Informatica recommends that you use an SSH connection without a password or have sshpass or Expect installed. If you do not use one of these methods, you must enter the password each time the utility downloads a file from the Hadoop cluster.

Verify the following host names: name node, JobTracker, and Hive client. Additionally, verify the locations for the following files on the Hadoop cluster:

- `hdfs-site.xml`
- `core-site.xml`
- `mapred-site.xml`
- `yarn-site.xml`
- `hive-site.xml`

Perform the following steps to configure Big Data Management:

1. Enter the name node host name.
2. Enter the SSH user ID.
3. Enter the password for the SSH user ID.

   If you use an SSH connection without a password, leave this field blank and press enter.
4. Enter the location for the `hdfs-site.xml` file on the Hadoop cluster.
5. Enter the location for the `core-site.xml` file on the Hadoop cluster.

   The Big Data Management Configuration Utility connects to the name node and downloads the following files: `hdfs-site.xml` and `core-site.xml`.
6. Enter the Yarn resource manager host name.

   Note: Yarn resource manager was formerly known as JobTracker.
7. Enter the SSH user ID.
8. Enter the password for the SSH user ID.

   If you use an SSH connection without a password, leave this field blank and press enter.
9. Enter the directory for the `mapred-site.xml` file on the Hadoop cluster.
10. Enter the directory for the `yarn-site.xml` file on the Hadoop cluster.

    The utility connects to the JobTracker and downloads the following files: `mapred-site.xml` and `yarn-site.xml`.
11. Enter the Hive client host name.
12. Enter the SSH user ID.
13. Enter the password for the SSH user ID.

    If you use an SSH connection without a password, leave this field blank and press enter.
14. Enter the directory for the `hive-site.xml` file on the Hadoop cluster.

The utility connects to the Hive client and downloads the following file: `hive-site.xml`.

15. Choose if you want to create connections to the Hadoop cluster:

Option	Description
1. Hadoop	Create a Hadoop connection.
2. Hive	Create a Hive connection.
3. HDFS	Create an HDFS connection.
4. HBase	Create an HBase connection.
5. Select all	Create all four types of connection.

Press the number that corresponds to your choice.

16. Supply information about the Informatica domain.
    a. Enter the domain name.
    b. Enter the node name.
    c. Enter the domain user name.
    d. Enter the domain password.
    e. Enter the Data Integration Service name.
    f. If the cluster is Kerberos-enabled, enter the following additional information:
        - Hadoop kerberos service principal name
        - Hadoop kerberos keytab location -- Location of the keytab on the Data Integration Service machine.

    Note: After you enter the Data Integration Service name, the utility tests the domain connection, and then recycles the Data Integration Service.

17. In the **Connection Details** section, provide the connection properties.

    Based on the type of connection you choose to create, the utility requires different properties. For more information about the connection properties, see the *Big Data Management User Guide*.

    Note: When you specify a directory path for the Blaze working directory or the Spark staging directory, you must specify existing directories. The Big Data management utility does not validate the directory paths that you specify.

    The utility creates the connections that you configured.

18. The utility reports a summary of its operations, including whether connection creation succeeded, and the location of utilty log files.
19. Complete the manual configuration steps for Big Data Management.

The utility creates the following files in the <Informatica installation directory>/tools/BDMUtil directory:

ClusterConfig.properties

Contains details about the properties fetched from the Hadoop cluster and connection creation commands that can be used to create connections to the Hadoop cluster.

Note: Edit the connection name, domain username and password to use the generated commands.

HiveServer2_EnvInfa.txt

Contains the list of environment variables and values that need to be copied to the HiveServer2 environment on the Hadoop cluster. This file is created only if you choose HiveServer2.

Verify Entry Files

After you run the configuration utility, verify that the files and entries are available in the required locations.

Verify that the ODBC entry files, TNS entry files, and DB2 installation entries are specified in the following location:

```
<Informatica installation directory>/services/shared/hadoop/<Hadoop_distribution_name>/infaConf/
hadoopEnv.properties
```

The following example shows the environment variables that you can edit:

```
infapdo.env.entry.ld_library_path=LD_LIBRARY_PATH=$HADOOP_NODE_INFA_HOME/services/shared/
bin:$HADOOP_NODE_INFA_HOME/DataTransformation/bin:/opt/teradata/client/14.10/tbuild/
lib64:/opt/teradata/client/14.10/odbc_64/lib:/databases/oracle_11.2.0/lib:/databases/
db2v9.5_64BIT/lib64:$HADOOP_NODE_HADOOP_DIST/lib/native:$HADOOP_NODE_INFA_HOME/ODBC7.1/lib:
$HADOOP_NODE_HADOOP_DIST/lib/native/Linux-amd64-64:$LD_LIBRARY_PATH

infapdo.env.entry.path=PATH=$HADOOP_NODE_HADOOP_DIST/scripts:$HADOOP_NODE_INFA_HOME/
services/shared/bin:/databases/oracle_11.2.0/bin:/databases/db2v9.5_64BIT/bin:
$HADOOP_NODE_INFA_HOME/ODBC7.1/bin:$PATH

infapdo.env.entry.oracle_home=ORACLE_HOME=/databases/oracle_11.2.0/
infapdo.env.entry.tns_admin=TNS_ADMIN=/opt/ora_tns
infapdo.env.entry.db2_home=DB2_HOME=/databases/db2v9.5_64BIT
infapdo.env.entry.db2instance=DB2INSTANCE=db95inst
infapdo.env.entry.db2codepage=DB2CODEPAGE="1208"
infapdo.env.entry.odbchome=ODBCHOME=$HADOOP_NODE_INFA_HOME/ODBC7.1
infapdo.env.entry.odbcini=ODBCINI=/opt/ODBCINI/odbc.ini
```

Copy the Highgroup.txt File to the Hadoop Machine

Before you can perform SSN masking, you must copy the `highgroup.txt` file from the Informatica services machine to the Hadoop machine.

1. On the Informatica services machine, browse to the following location: `<Informatica installation directory>/server/infa_shared/SrcFiles`

   The path indicates the default location of the file.
2. Copy the `highgroup.txt` file to a location on the Hadoop machine.

   Copy the file to all nodes in a cluster. The path to the file must be the same on all nodes.

Import Connections in to Test Data Manager

You can import the connections that the configuration utility creates in to Test Data Manager.

1. Log in to Test Data Manager as an administrator.
2. In the **Administrator | Connections** view, select **Import Connections** from the **Actions** menu.
3. The available source and target connections are imported into Test Data Manager from Informatica service.
4. Go to the **Administrator | Connections** view to see a list of imported connections.

   The connections that you created with the configuration utility are visible in the list of connections.

Create Connections in Test Data Manager

You can create a source or target connection in Test Data Manager.

1. Log in to Test Data Manager as an administrator.
2. In the **Administrator | Connections** view, select **New Connection** from the **Actions** menu.

A tab opens to display the new connection properties.

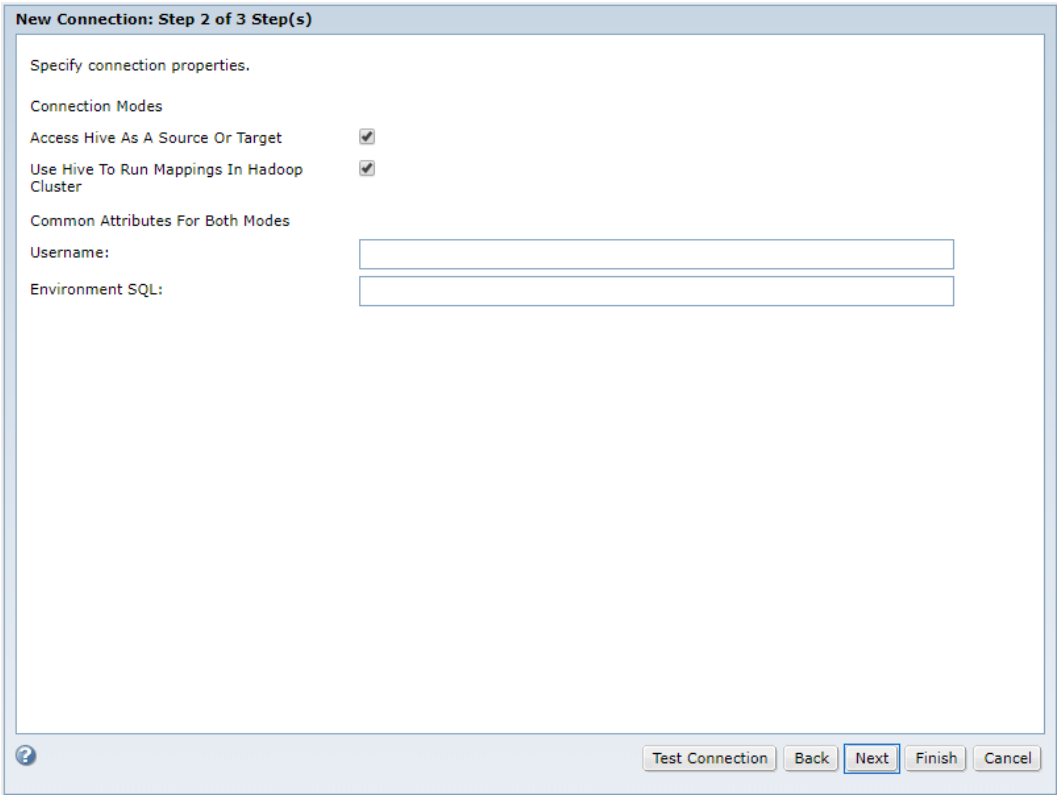
3. Select the connection type and define the connection name, description, and user information.

   You can create a Hive, Hadoop, or HDFS connection. The connection name must begin with an alphabetic character. If you enter a connection name that begins with a numeric character, a workflow that includes the connection might fail.

4. Click **Next**.

5. Enter the connection properties. The number of steps and the properties that appear differ based on the type of connection you select.

   For information about connection properties for each of the connection types, see the *Test Data Management 10.1.0 Administrator Guide*.
   The following image shows the second step of the Hive connection creation wizard:



6. Optional. Click **Test Connection** to test the connection.

7. Click **Finish** to save the connection.

8. Optional. Highlight the connection in the **Administrator** | **Connections** view, and click **Actions** > **Test Connection** to validate the connection.

You can use the connection in a plan.

Configure Pushdown Connection Properties in Test Data Manager

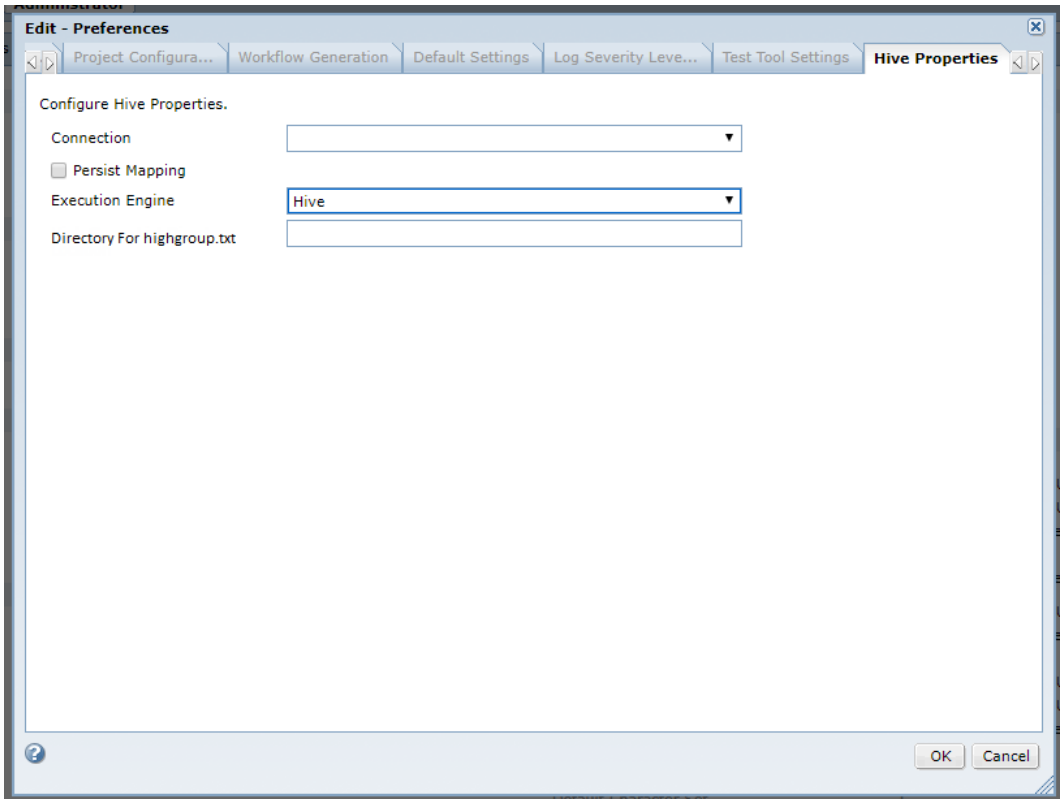
Workflows that contain Hive or HDFS connections must be able to run mappings from Test Data Manager. You must configure a Hadoop connection before you can run mappings in Test Data Manager.

1. Log in to Test Data Manager as an administrator.

2. Click **Administrator** > **Preferences**.

3. In the **Hive Properties** section, click **Edit**.
4. Select the Hadoop connection from the list.
5. To view the mappings that the Data Integration Service runs in the Administrator tool, enable Persist Mapping.

   The following image shows the **Hive Properties** tab in the **Preferences** window:



6. Select the required execution engine to run the mappings. You can choose Hive or Blaze.
7. Enter the path to the `highgroup.txt` file on the Hadoop machine.

   Required if you want to use the SSN masking rule.
8. Click **OK**.

Create a Project and Import Source Metadata

Create a project and import the metadata into the project to perform TDM operations on source data.

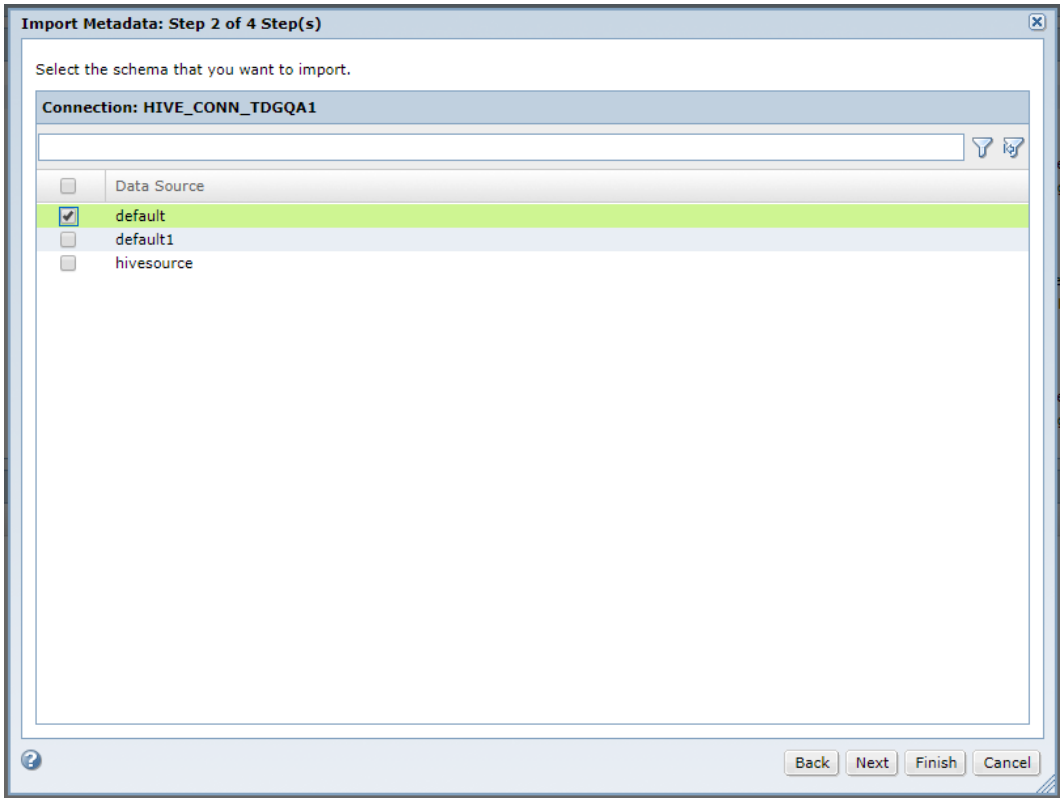
1. Log in to Test Data Manager.
2. Click **Projects** to access the projects.
3. Click **Actions** > **New**.

   The **Create Project** dialog box opens.
4. Enter a name and optional description and click **OK** to create the project.
5. Click **Actions** > **Import Metadata**.

   The **Import Metadata** window appears.
6. Choose to import metadata from a data source connection, select the required Hive connection from the list, and click **Next**.

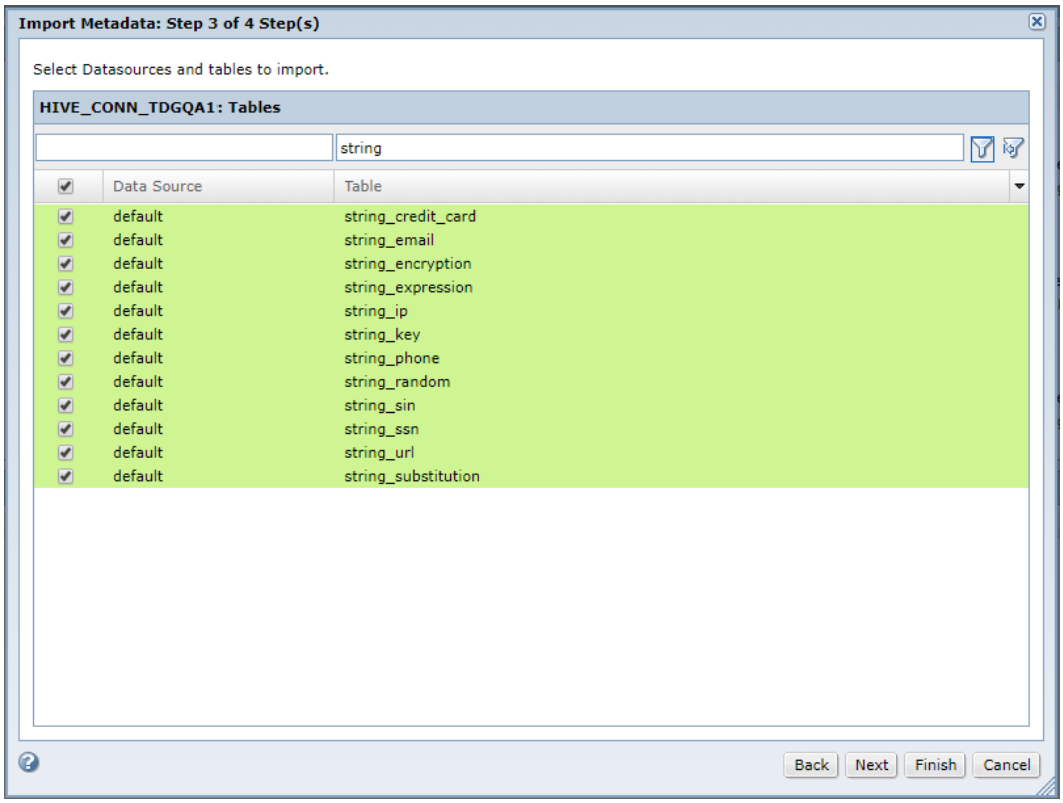
7. Select the schema that contains the source metadata and click **Next**.

   You can filter to search for the required schema.
   The following image shows the schema selection window in the **Import Metadata** wizard:



8. Filter and select the data sources and tables that you want to import and click **Next**.

The following image shows the table selection window in the **Import Metadata** wizard:



9. Choose to import the metadata now and click **Finish**.

You can use the source metadata from the connection to perform TDM operations in Test Data Manager.

Author

Sadhana Kamath

Acknowledgements

The author would like to acknowledge Krishnakanth K S, for his technical assistance.